

VaultlineDB — System PRD & End-to-End Flow

Digital Banking Wallet Engine Architecture & Project Explanation Document

VERSION 2.0 • COMPLETE SYSTEM OVERVIEW

1. Quick Elevator Pitch (Project Explanation)

Project Concept:

VaultlineDB is a secure digital banking and wallet transactional backend system (similar to JazzCash, Easypaisa, or NayaPay). It manages multi-role users (Customer, Merchant, Admin), handles real-time peer-to-peer and QR merchant transfers with ACID compliance, and executes automated security lockouts and high-value fraud detection.

2. User Roles & Dashboard Breakdown

1. Customer Dashboard (End User)

- **Core Purpose:** Peer-to-Peer money transfers, balance management, and merchant payments via QR code scanning.
- **Daily Limits:** Standard threshold of PKR 50,000.00 per day.
- **Technical Execution:** Connects to `sp_PerformTransfer` procedure for fund movements.

2. Merchant Dashboard (Business Owner)

- **Core Purpose:** Receiving payments from customers and business financial management.
- **QR Code Mechanism:** Displays a unique static/dynamic QR code mapped directly to the Merchant's `AccountNo` (e.g., `VL-10492`). Customer scans the code to fetch destination details and execute transfer.
- **Daily Limits:** Higher transactional allowance of PKR 200,000.00 per day.

3. Admin Dashboard (System Controller)

- **Core Purpose:** Security monitoring, fraud auditing, and system control.
- **Automated Rules:** Monitors account lockouts (`sp_HandleFailedLogin` after 3 failed tries) and receives real-time alerts from `trg_CheckHighValueTransaction` for transfers $\geq$ PKR 100,000.00.

3. End-to-End Technical Flow

STEP 1 User Registration & Dynamic Wallet Allocation

Execution of `sp_RegisterUser` inserts profile into `USERS`, checks role (Customer vs. Merchant), sets custom daily limit, and automatically provisions a linked wallet in `WALLETS` table using `SCOPE_IDENTITY()`.

STEP 2 Money Transfer & QR Code Payment Processing

Execution of `sp_PerformTransfer` validates sender balance, opens a `BEGIN TRANSACTION` block, deducts sender balance, credits receiver balance, logs entry in `TRANSACTIONS`, and executes `COMMIT` (or `ROLLBACK` on failure).

STEP 3 Security & Automated Account Lockout

Execution of `sp_HandleFailedLogin` increments `FailedLoginCount`. If invalid attempts reach 3 (`MaxFailedLogins`), status is set to 'Locked' and logged in `SecurityLogs`.

STEP 4 Automated Fraud Alerting (Trigger)

The trigger `trg_CheckHighValueTransaction` automatically intercepts every insert in `TRANSACTIONS`. If transfer amount $\geq$ PKR 100,000.00, it flags a high-risk record in `FraudAlerts`.

4. Database Core Schema Summary

Table	Primary Key	Key Foreign Keys / Columns	Role / Description
USERS	UserId (1001+)	Email, Role, Status, FailedLoginCount	User profile & security credentials
WALLETS	WalletId (5001+)	UserID (FK), AccountNo, Balance, DailyLimit	Financial accounts & balances
TRANSACTIONS	TransactionId (88001+)	SenderWalletId (FK), ReceiverWalletId (FK)	Transfer & payment ledger
FraudAlerts	AlertID (1+)	TransactionID (FK), UserID (FK), RiskLevel	High-value transaction audit flags
SecurityLogs	LogID (1+)	UserID (FK), ActionType, Description	System audit trail for security events
SystemSettings	SettingID (1+)	MaxDailyLimit, HighValueThreshold, MaxFailedLogins	Global configuration thresholds